TaMoS: Example questions and student answers

These questions have been asked in previous exams during 2020/2021 and the following answers have all been graded with 5/5 points (part 2) or A (part 3). Please note that this does **not** mean that these answers are necessarily beyond improvement nor that they can be used instead of the textbook. You are not allowed to copy from these texts into your own answers: that will be considered plagiarism. These answers have been lightly edited. One reason for this is to improve clarity, e.g. correcting spelling, phrasing and removing passages that the student included, but which were not asked for in the question and thus not graded. Another reason is to preserve anonymity. One final comment: most of these students have answered in more detail than was required. There is no "minimum length" for these questions.

Part 2.

Example 1.

First, describe, in general terms, what an operationalization is. Account for two of the main reasons for operationalizing a phenomenon or property and illustrate both with the help of examples.

Next, account for the thesis of operationalism and give at least one argument against this thesis.

Finally, discuss the difference between operationalizing a concept or property and giving a stipulative definition of this concept or property.

Answer 1.

Description of operationalization

In some cases, it is not possible to directly observe a phenomenon, not even with aids. Rather, the property of interest has some effect that we instead can directly observe. Thus, the property of interest is linked to a directly observable effect through a hypothesised causal chain. Extra emphasis should be put on the term "hypothesised", as operationalizations are theory dependent. That is, we need theory in order to justify the operationalization. An example from social science is that we cannot directly observe a person's thoughts. We use a questionnaire or an interview from which we can directly observe the subject's written or verbal response to our questions. The underlying hypothesis here is that interview or questionnaire adequately captures the subject's mental attitude.

Even if we can measure something directly, we often often prefer to measure indirectly through an operationalization. There are a couple of reasons for this. The first reason is that direct observations are often unreliable. The second reason is that direct observations typically only account for qualitative values. These reasons will be shown in the following example.

Take temperature for example. We do not directly observe mean kinetic energy, but we use for instance a thermometer for this. What we then directly observe is how high the fluid in the thermometer has risen. What lies behind this is the hypothesis that there is a linear relationship between the liquid's density and temperature, and the height at which the fluid has expanded in the column. If we would measure this directly, we could not give an accurate value of the temperature as we could with a thermometer. We could for instance probably say that the water is "warm", but not 32,5 degrees Celsius. If we would be guess a temperature, we would never be as accurate as the thermometer.

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Operationalism

Operationalization should not be confused with operationalism, which is a controversial stance on measurement. Operationalism is a view that describes that the properties of interest that we measure are defined by the very operations by which they are measured. Going back to the social sciences, let us say that we want to measure "happiness" in a person. We operationalise happiness through the appearance of a smile on our subject. The operationalist would argue then that happiness is the smile.

It is clear that there are problems with this stance. First of all, we multiply concepts that should not be multiplied. If we for instance measure length in different ways, the operationalist would argue that these are different concepts of length, which does not seem correct. Another disadvantage is that operationalism make sus lose the ability to critically evaluate the operationalisations that we make in science. If we indeed multiply concepts like previously described, then the operationalist could point to that all the concepts are different. Thus, he can argue that they should not be compared, and thus escapes the critique on his operationalization. This is very serious, as critical peer reviewing is one of the most important pillars of science.

Differences between operationalization and stipulative definition

Stipulative definitions are definitions that are made for a specific purpose. Therefore, they can be said to be purpose dependent. That is, the definition can be evaluated by how useful it is given a certain context. It can be said that there are mainly two differences between providing operationalizations and stipulative definitions of a concept or property.

First, operationalizations can never be stipulative. One cannot simply stipulate the causal relationship between the property of interest and the observed effect. Unlike stipulative definitions, the question of whether an operationalization is valid is an empirical question. For stipulative definitions, the question of validity is merely purpose-dependent.

Second, the links between the property of interest and the observable effect is different from the connection between the definiendum and the definiens. That is, there is a logical connection between the definiendum and the definiens. For operationalizations, the connection is weaker in the sense that one proposes a connection between the property of interest and the measurable effects.

One does not identify the property of interest with the observable effects.

Answer 2

An operationalization is an indirect observational process. If we operationalize a property of interest, we give a method to link it to a directly observable effect using a hypothesized causal chain or a correlational chain. Generally speaking, operationalization is a method through which we can measure something or a property of interest which cannot be observed directly or that can be observed directly but NOT with sufficient precision, by linking it to something that can be observed directly. However, we need to be aware that not every method or property results in a good operationalization.

Operationalization is used when the property of interest:

1. Is not directly observable.

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Example: we would like to find the refractive index of a medium. Here, the property of interest is Refractive Index.

We know, from Snell's law that the ration of sine of angle of incidence to the angle of refraction is equal to the ratio of velocities of light in the two media (medium 2/medium 1). Also, the ratio of velocities is inversely proportional to the ratio of refractive indices of the two media (medium 1/medium 2). This acts as a stable relation.

Velocity of light in different media can be measured and given that we know the velocities, we can calculate the refractive index of one medium with respect to another (where, the reference medium is usually taken as air). This is our desired effect or property.

2. Directly observable but not with sufficient precision.

Example: We are interested in finding the Hall Voltage produced in a sensor under the effect of certain electric and magnetic field.

In Hall Effect, a potential difference is produced across an electrical conductor, in a direction perpendicular to the flow of the electric current in the conductor and to an applied magnetic field that is also perpendicular to the direction of the flow of current and is proportional to the magnitude of the current flow and the magnetic field strength. This is the stable relationship.

The voltage produced by this phenomenon is our desired effect. This example does require sufficient precision in measurement of the voltage as Hall Effect sensors are often used to regulate the magnetic field strength in a Cyclotron by a feedback that continuously monitors the Hall Voltage.

Operationalism is a view that tells us what an Operationalization can do and should do but in most cases, it fails to align itself with the criteria of a good operationalization. Operationalism is the domain where we believe that most of the concepts are defined through the operations that are used to observe or measure them. Operationalism eliminates the theory-dependence factor from the measurement process. Adopting operationalism inadvertently makes us lose the important ability to evaluate an operationalization critically. Once the theory dependence is removed from a certain observational process, it becomes very hard to evaluate the process of measurement and the operationalization as a whole.

Operationalizing a concept or property means that we give a method or set of procedures to observe or measure that concept or property whereas a Stipulative definition is a DEFINITION, i.e. we only define the relevant concept or property. A stipulative definition would only allow someone to know about the concept or property as someone else wanted them to know but Operationalizing would give a person the means to measure that desired effect or property through an experimental setup or other methods. The operationalization may or may not produce the desired results whereas and similarly, the stipulative definition may be vague or too broad with respect to the context.

Example 2.

Explain the concept of a neutral analogy in modelling, and discuss what is meant by the claim that a neutral analogy holds between a model bridge in a computer simulation and a real bridge with regards to the property "ability to withstand an earth quake".

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Finally, account for one positive analogy and one negative analogy in relation to this model and target. Discuss if a scientist should aim to change these negative analogies into positive analogies or not, and what relationship positive and negative analogies have with the external validity of the inferences drawn from the model.

Answer 1.

First neutral analogy is a concept that was introduced by Mary Hesse in 1963. An analogy is a property that is compared between the model and the real target for example the size, the robustness or the ability to resist to something. If the scientist knows that this property occurs in both the model and the real target then it is a positive analogy. Otherwise, if this property lacks in one of them but occurs in the other then it is a negative analogy. Finally, if one is not certain about the fact that the target has the property or not then, this analogy is classified as neutral.

If a neutral analogy holds between a model bridge in a computer simulation and a real bridge with regards to the property "ability to withstand an earth quake", then it means that it is not certain that the bridge is able to withstand an earth quake. An example of a positive analogy between the model bridge and the real bridge is for example that their deformations are governed by the equations of solid mechanics. A negative analogy might be that for simplicity reasons the wind has not been taken into account in the model and thus no wind interact with the model bridge whereas wind is actually generating a force on the real bridge.

The scientist can turn this negative analogy into a positive one to increase precision but this will decrease the simplicity of the model drastically. Moreover, introducing Navier Stokes equations into the model to takes into account the wind will also decrease the transparency of the model. Such equations are really hard to understand and to use so the model user might not be able to really understand how the result is processed. Concerning external validity, turning this analogy into a positive one will of course increase the external validity depending on the impact the wind has on the real bridge. I would say that if the wind has a big impact and for example induce a resonance in the bridge during the earthquake there is a high chance that the model could give a wrong result. However if the bridge is shaped in a way that the wind flows around him without creating too high forces on it, the external validity is then not really impacted.

Answer 2.

First, the meaning of we should define what does analogy mean in modelling. To infer by an analogy is that if there are similarities between two cases, then a conclusion that applies for one of the cases, applies for the other.

An inference analogy starts by stating that a model and a target are similar in some aspects (therefore share positive analogies), and not in others (and therefore share negative analogies). Then further properties P of the models, which are not known if the target possesses them, are to be identified. Those are neutral analogies. The inference ends by concluding, through the fact that the model and target share similarities, that the target also possesses the properties P.

To effectively conclude by analogy, the neutral analogies must causally arise from the positive analogies' properties of the model. We can apply this way of thinking to bridges. Let us suppose that the target is a real bridge, and that the model is the same bridge, but in a computer simulation. What is meant by "a neutral analogy holds between a model bridge in a computer simulation and a real bridge with regards to the property "ability to withstand an earth quake" " is that the model bridge can withstand an earthquake in the computer simulation, but we do not know if the real bridge would

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withstand it in reality. One positive analogy would be that the real bridge and the model bridge have the same shape. One negative analogy would be that the model could not represent all real-world variables, like how aging will affect the bridge for example. The negative analogy would be "while the model bridge will always be the same, the real bridge will change as years go by". A scientist should aim to change this negative analogy in a positive analogy, however, simulating real world aging of the bridge seems difficult due to all background variables of the real world. If this negative analogy is successfully transformed into a positive analogy, then it can boost the confirmation in the inference that the real-world bridge will have the "ability to withstand an earth quake".

Negative and positive analogies affect the external validity of the simulation. Let us say that a hypothetical model shares a lot of positive analogies with a target with only one negative analogy. Even so, how can we be sure that the negative analogy does not affect the external validity of the inference? For a simulation to be externally valid, one has to carefully check if the negative analogies are outweighing the positive analogies. Let us return to our example of the bridge. If the negative analogy is "the model is blue while the target is red", then we can confidently say that it does not affect the external validity of the "ability to withstand an earth quake" inference. However, if the negative analogy is "the model bridge is simulated in wood and the real one is in iron", then we would not be sure that the inference is externally valid.

Example 3.

First, distinguish between deterministic hypotheses and stochastic hypotheses, and give one example of each in relation to investigating the connection between train delays and people walking on the railroad tracks. Also, explain why statistical methods are necessary for quantifying measurement error when evaluating hypotheses.

Next, explain the concepts "test statistic" and "p-value" in the context of Fisher's statistical hypothesis testing. Further, explain the relationship between the test statistic and the p-value in a case where hypothesis H is rejected based on an experiment.

Finally, explain one advantage with using Bayesian hypothesis testing, in comparison with Neyman-Pearson's or Fisher's methods.

Answer 1.

Deterministic hypotheses make absolute claims about a set of outcomes. By contrast, stochastic hypotheses make claims about the probability of a set of outcomes. As an example, consider the connection between train delays and people walking on the railroad. A deterministic hypothesis about this connection might be "If someone walks on a railroad while a train is travelling on it, that train will be delayed," whereas a probabilistic hypothesis might be "During the time that people walk on a railroad, the probability that a train travelling on this railroad is delayed doubles."

Consider a scenario in which one has some hypothesis involving a claim about a quantity that must be measured. Furthermore, suppose that this measurement has the possibility of error. If a measurement that conflicts with the hypothesis is made, an important question arises: Is this due to the hypothesis being wrong, or a consequence of an erroneous measurement? Statistics are vital in answering this question, and its role is, paraphrasing the course literature [1], to provide quantification of the measurement error and associated probabilities. For instance, if there is a low probability that the true value of the measurement is within the range allowed by the hypothesis, one might still be justified in rejecting the hypothesis.

Consider an experiment with a set of possible outcomes and start with some hypothesis H that probabilistically describes the outcomes of this experiment. The test statistic is the set of possible outcomes of the experiment and their respective probabilities. The p-

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value is the probability that the outcome is at least as extreme as the observed outcome, given H. In a particular case, once an outcome is obtained, one can compare this to the test statistic. Summing up the probabilities of all outcomes at least as extreme as the one observed would yield the p-value. In a case where H is rejected, the p-value would be found to be higher than a previously set significance level.

One advantage of Bayesian hypothesis testing when compared to the Fisher- or Neyman/Pearson schemes lies in its handling of hypotheses for which counter-evidence is found. In the latter schemes, the discovery of counter-evidence might lead to the outright rejection of a hypothesis. In the Bayesian scheme, however, the probabilities of different hypotheses being true (given previous observations) are continuously updated. In the case of counter-evidence, one would find that the probability of hypotheses conflicting with this observation is lowered. This allows for handling and considering multiple hypotheses at a time.

1. Grüne-Yanoff, T. Experiments, Models and Methodology Nov. 2020. [comment: here a page number would be ideal.]

Answer 2

Deterministic Hypothesis - The deterministic hypothesis is quite absolute and says that if we have implications of an outcome then this outcome is certain, it either happens or it doesn't.

In the case of train delays and people walking on railroad tracks, a deterministic hypothesis would say that: "The trains are delayed when people are walking on the tracks". In this hypothesis we can find certain implications that when someone is walking on the rails, the trains will be delayed as a result.

Stochastic Hypothesis - The difference between stochastic and deterministic hypothesis is that the stochastic hypothesis relies on probability distributions. It doesn't say for certain that something will happen or does happen, but rather it tells us the probability of what outcome we will get.

Applying this to the train delay example, a stochastic hypothesis would have to have an outcome where we can't be fully certain about the implications, for instance: "People walking on the railroad tracks cause train delays 70% of the time". Here we would have to look at many cases of train delays to start seeing patterns trending toward 70% or not. Of course, we will never reach complete certainty seeing as that would require an infinite amount of cases to be studied.

Statistical methods are necessary since if one measure or test seems "off" or rather seems to indicate that we should reject our hypothesis we have to make sure it wasn't just an outlier/measurement error. If we perform the test many times we can start to notice patterns in the outcomes. We can avoid having to falsify our observation if we take measurement error into account.

In Fisher's statistical hypothesis testing the central part is a significance test. This is done by defining the *test statistic, which* constitutes all the possible outcomes and the respective probability of said outcomes of the test. This basically means that we have a theoretically calculated case to compare to the case of our actual outcome of the test. It is a step on the way and helps us to determine whether we're far off enough to reject a hypothesis H.

The so-called *p-value* is the value used to determine whether the probability of an observed outcome of a hypothesis is low enough to reject said hypothesis by comparing it to a set conventional standard also known as a *significance level*. If the p-

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value is lower than the significance level, we reject hypothesis H. In Fisher's method, the p-value is determined by taking the sum of the entries in the sampling distribution of the test statistic that correspond to the probability of observing an outcome like the one we just observed (or more "extreme" as the course literature puts it).

Using Bayesian hypothesis testing over Neyman-Pearson's or Fisher's methods gives you the ability to determine the degree of confidence in the hypothesis being tested. Bayesian statistics does this by looking at the prior probability of the hypothesis, then applying Bayes theorem on these probabilities and then obtaining the posterior probability. Neyman-Pearson's or Fisher's methods are quite absolute and tell us to either reject or not reject a hypothesis, but provide no such tools to handle degrees of confidence.

Example 4.

Explain the notion of belief-desire explanation, and provide an explanans for the explanandum: "Why did Anne lie and say that she had not bought her son any presents, instead of telling him the truth?" Also, give an example of a method to determine if this explanation is the correct one.

Next, distinguish qualitative data from quantitative data, and discuss whether the data produced by the method you stated in the previous paragraph is qualitative or quantitative.

Finally, provide two quality criteria for qualitative methods and explain in what way they improve the quality of an operationalization.

Answer 1.

The notion of belief-desire explanation is to provide an explanation why an agent A acted a certain way, rather than another way, by attributing a belief and desire in the agent. By saying that the agent wanted a certain result and believed that action would bring about that result, explains the action.

Now, we will provide an explanans for the explanandum "Why did Anne lie and say that she had not bought her son any presents, instead of telling him the truth?". One possible explanans is Anne desires her son to be happy. Anne believes that surprising her son and lying would bring this result. A method to determine if this explanation is true is to interview Anne and ask question that would generate data to help us interpret her believes and desires.

Quantitative data is data that involves numbers while qualitative data is data that does not involve numbers, at least directly. Qualitative data is often involve comparisons or interpretations and is featured in reports and interviews. However, qualitative data may of course often be quantified. The data generated in the aforementioned method, is qualitative because it involves an interview, which is not numerical. Some of the data may however be summarized in quantitative data.

There are quality criteria for qualitative methods that improve the quality of an operationalization. It should be noted that two sets of strategies are often identified in this context, either to avoid interpretation all together (e.g. Behaviourism) or to improve interpretation. By identifying quality criteria as specific strategies or properties of qualitative methods, we can provide two such examples. In the context of interpretation improvement, we have attention checks and repeat questions in surveys. An attention check is the strategy to check if a survey respondent actively answers the question (i.e. not randomly selecting answers) and improves the operationalization by making sure that survey answers actually reflects reality and not a random process. Repeat questions asks the same question in a survey that improves operationalization by making sure that the question accurately describes the feature of interest and that the user has paid attention.

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Answer 2.

A belief-desire explanation explains an action by the belief and desire of the person taking the action. The person desires some result, and believes that taking the action leads to that result. Annie desired to give a positive surprise to her son. She believed that lying about not having bought him presents when she actually had would lead to a positive surprise once she revealed her lie.

To determine if the explanation was correct, we could interview Anne, asking questions related to her beliefs and desires, like: Did you think that lying would result in a positive surprise? Did you want a positive surprise for your son? Using several competent observers to reduce interpretation error, and repeat questions to make sure Anne is paying attention in the interviews.

Qualitative data is data that cannot be counted or statistically compared. Quantitative data is the opposite, where different sets of data are directly comparable. The chosen method of interviewing produces qualitative data in the form of open interview answers, which the observers need to interpret. The interpretations could be later quantified by assigning numbers for the observers' answers (0 – explanation is correct, 1 – explanation is incorrect) and taking the average.

Procedural rigour: Means to show how the data was collected and analysed. This makes the method repeatable so that the results could be checked. It improves the operationalization by providing a possibility to evaluate the validity of it.

Triangulation: Means to use more than one observation method to analyse a feature of interest. This improves the operationalization by revealing or mitigating effects of biases related to the individual methods. If several operationalisations yield the same result, we may increase trust in them.

Part 3.

Example 1.

One strength the deductive-nomological account (D-N account) of explanation is that it is deductive. Explain what it means that the D-N account is deductive, and why this property is considered a strength of the D-N account.

Some theorists are of the opinion that scientists should rely on deduction as much as possible (and, by the same token, avoid non-deductive inferences to the furthest extent possible). Account for one argument in favor of this view as well as one argument against it.

Now, consider the causal account of explanation. The causal account and the D-N account can both overlap with each other as well as diverge from each other. Give an example of a causal explanation and a D-N explanation that do overlap in the relevant sense, and an example of a causal explanation and a D-N explanation that do not.

Answer 1

"One strength the deductive-nomological account (D-N account) of explanation is that it is deductive. Explain what it means that the D-N account is deductive, and why this property is considered a strength of the D-N account."

Deductive inferences are inferences that explicate knowledge that we already know. Thus, they are truth-preserving, meaning that if the conditions are true, the conclusion must necessarily also be true. Therefore, deductive inferences are not fallible, as opposed to inductive inferences. Inductive inferences amplify knowledge by going

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beyond what we already know. An inductive inference can be wrong, even if the conditions are true.

The deductive-nomological (DN) account describes that the explanation of a phenomenon, the explanandum, is if one has reasons to expect the phenomenon based on natural laws or law-like generalizations combined with a set of initial conditions. It is deductive, for an explanandum is explained by *deducing* it from a set of laws or law-like generalizations, the explanantia. According to the DN, one can explain a phenomenon if one can explicate already acquired laws or law-like generalizations to conclude that the phenomenon must happen.

This property is considered a strength, because, as mentioned above, deductive inferences are infallible. This means that if the laws or law-like generalizations as well as the initial conditions are all true, there is no way that the phenomenon will not happen. This is, of course, a rather strong claim and a rather reliable way of understanding and explaining a phenomenon. If one uses the DN account, one's explanations of a phenomenon are infallible, they are never wrong, unless one makes a mistake regarding the laws or the initial conditions, of course. This can furthermore be used for falsification. If a phenomenon that should be expected based on the DN account does not occur, and there is no error in the initial conditions, one can assume that there is an error in the laws or law-like generalization and thus falsify them. In reality, this can be complicated by the Duhem-Quine thesis, making it impossible to know if the error came from a main hypothesis or an auxiliary hypothesis such as operationalizations used for the observation, but the argument is still valid to a certain degree.

"Some theorists are of the opinion that scientists should rely on deduction as much as possible (and, by the same token, avoid non-deductive inferences to the furthest extent possible). Account for one argument in favor of this view as well as one argument against it."

An important argument against non-deductive inferences follows from Hume's problem of induction. Hume states that every inference is either an induction or deduction. Due to the fallibility of inductive inferences, and inductive inference rule I has to be justified by inferring it from somewhere. However, this inference cannot be of deductive sort, as Hume states that there is no necessary connection between past and future inferences. Hume concludes that the inference rule I therefore has to be inferred inductively. However, this inductive inference requires justification again, leading to an infinite regress of inductive inferences. Ultimately, this means that no inductive inference can be truly justified, as "justification" is only ever provided by an infinite set of equally unjustified inference rules. One can therefore argue that non-deductive inferences are to be avoided, as they are never justified. A deductive inference rule, however, preserves truth and requires no justification beyond the original rules it was based upon.

An argument against the disastrous implications of Hume's problem of induction can be that it assumes science to require some sort of foundation, a ground truth to base one's inferences on. This notion is referred to as foundationalism. However, one can argue, there is no such core truth in science. Instead, real-world science usually tries to create a coherent system of relations and explanations, and a claim is considered justified if it is compatible with the existing system. Metaphorically, this can be compared to a boat on the ocean that is being repaired and maintained by its crew while being afloat, rather than placing it in a drydock to conduct these works. According to this view, coherentism, rather than creating an infinite regress of induction based on nothing, Hume's problem leads to a long circle of inductions which is considered enough to create a coherent system that can "stay afloat" by itself.

Having "defused" Hume's problem, one can make an obvious argument for the use of inductive inferences in science. Inductive inferences are common scientific practice, and they are often a straightforward or even the only way to justify new hypotheses and generate knowledge. A scientific system that relies solely on deductive inferences would never be able to consider a theory as right, rather it will only ever see it as non-falsified

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(compare this to Popper's falsificationism), as there is no way to inductively prove a theory right, while there is (at least in theory, remember the Duhem- Quine thesis) a way to deductively prove a theory wrong. The scientific progress of today, which has arguably led to a lot of advancements in human history, relied heavily on the use of inductive inferences. Therefore, one should embrace inductive inference rules in science, as long as they are justified in a coherent set of other explanations.

"Now, consider the causal account of explanation. The causal account and the D-N account can both overlap with each other as well as diverge from each other. Give an example of a causal explanation and a D-N explanation that do overlap in the relevant sense, and an example of a causal explanation and a D-N explanation that do not."

The causal account is when an explanation of a phenomenon, the explanandum, is made by stating causes as explanans. Causes are factors that have a difference-making influence on the explanandum. According to the causal account, it is sufficient to explain "Why is it so dark in this room" with "Because the lights are turned off" rather than by stating a law or law-like generalizations as explanans, as would be the case in the DN account. If the lights were not turned off, it would be brighter in the room, so this explanans is a cause. A different causal explanation could be "Because it is night time", since, if it were daytime, the light coming through windows would brighten up the room, too. The causal account allows for these differing explanations. To narrow down the amount of competing explanations, one can specify a contrast in the explanandum, asking for a specific difference-making factor, e.g. "Why is it so dark in this room now, but it was bright 5 hours ago.".

One can construct a rather simple example on which to show both, an example of overlapping and diverging DN and causal account. Take for instance a flagpole of length x that has a shadow of a certain length y on a sunny day.

As an example of overlapping explanations, one could want to explain why the shadow has the length y, a *DN explanation* could be "Because an object of length X on a sunny day where the sun is at an angle Phi has a shadow of length Y=X*cos(Phi)/sin(Phi)" as law and "the flagpole is x long and the sun is at angle phi" as conditions. Here, one cites a physical law in the form of a mathematical equation. This law generally applies and deductively explains the length of the shadow. Similarly, a *causal explanation* could be "Because the flagpole is x long and therefore casts a shadow of length y." Again, one explains the length of the shadow by the length of the flagpole, so the two explanations overlap.

The counterexample uses the same setup, but this time concerns a different explanandum. If one investigates the opposite direction and asks "Why is the flag pole x long", a valid explanation *according to the DN account* would be to reverse the above statement and rearrange the above formula to X=Y*sin(Phi)/cos(Phi) and consequently explain the flagpoles length by the length of its shadow. This seems very counterintuitive, as our understanding of causation is asymmetric, and an object in the sun casts a shadow, but not the other way around. A *causal explanation* could, in contrast, be "Because the client specified it to be this length". This states a difference-making cause, since if the client had ordered a different length, the flagpole would be longer or shorter, and is thus a causal explanation. The shadow, however, is not a different making cause for the causal account since even if the shadow was longer, the flagpole would still be the same size. Explaining the flagpoles length by the length of its shadow is therefore not possible using the causal account. The other way around it is also impossible to use "Because the client specified it to be this length" as an explanation in the DN account, as it does not represent a law or law-like generalization.

Example 2.

Consider the following case. You want to study the effects on people's social life after starting to use dating-apps. For your study, you will instruct the subjects to use a certain dating-app, which standardly collects information about the users: data that you will use for your study.

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Now, if you are to perform this study with informed consent, is it sufficient that the users sign the standard user-agreement (the same one that every user agrees to, the "terms and conditions" or end user license agreement) to use the application? Explain the notion of informed consent and argue whether the standard user-agreement is sufficient. Next, discuss and justify which qualitative methods could be employed to determine whether you have succeeded in using informed consent or not. Take a position on whether you have problems with the quality of your operationalization(s), and present arguments in favor of your view.

Answer 1

Def.1 Informed Consent: Informed, voluntary, decision-capacitated consent.

To use the signature on the user-agreement as proof of informed consent presents multiple problems.

The first problematic aspect refers to the informed consent on the app usage. A user-agreement presents information mainly regarding user data management in the hands of the company behind the app. To sign up to a service such as the one provided by a dating app, a user is required to give to the software a number of personal information which could be processed by the company for instance to improve personal advertisement algorithms. This poses a big problem on privacy that is usually avoided thanks to the user-agreement, which should inform the participant on all the relevant aspects of the data managing. This however does not satisfy all the requirements for informed consent. In particular, even if the user is theoretically well informed and the consent is deemed voluntarily, one can not be guaranteed that the consent was decision-capacitated. In particular, the information provided could be highly technical and the user could not understand all of it or simply skip it (a common practice among users of internet services). Services usually do not verify the correct understanding of information, therefore the user agreement lacks some of the necessary requirements of informed consent by itself.

If this wasn't enough, the second problematic aspect arises in terms of using an app-specific user agreement as proof of informed consent for participation in an experiment. In particular, even if the consent remains voluntary, the requirement on information is unlikely to be satisfied, as it is highly unlikely a user agreement possesses information on the relevant aspects of a specific experiment unless the app was specifically developed for such an experiment. Information on the experiment could include for instance data handling by the scientists, explanation on the methods of observation (if not limited to data analysis), and possible psychological consequences of the exposure to dating-apps (especially if the subjects are new to the use of dating-apps).

Qualitative methods for quality evaluation of each of the requirements for informed consent are analysed as follows.

Informed: To understand if experimenters gave all the relevant information about the study to participants, such information could be exposed to a third-party commission, which will give approval or rejection on the quality of the information. The construct validity of such an operationalization is based on the hypothesis that such a commission would be neutral (it can't be part of a dating-app company for instance) and that it would understand all of the information given to it (this depends mainly on the collective expertise of the individuals forming the committee). Since such a commission is deemed to exist (for instance, it could be an academic commission formed by

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researchers without conflicts of interest), the operationalization is considered to be satisfying all quality criteria.

Voluntary: A simple signature could be used to test if the consent is voluntary. This however poses a problem, as subjects could be exposed to external pressures in signing up to the experiment. The subject for instance could be signing for economical reasons. This could be accounted for by analysing data on the financial background of the subject and selecting only those who reach a certain annual income threshold. Of course, this would for instance exclude students who are being financially aid by their parents and therefore be able to participate to the study voluntary even without reaching such a threshold. Moreover, the threshold could be modified ad hoc to gain participants. To solve this issue, this method too could be exposed to a third-party commission, accounting for all the issues stated above.

Decision-capacitated: To ensure the subject is decision-capacitated is first to ensure he/she has understood all the relevant information that was handed to him/her. One way of doing this would be to simply asking it, with a yes/no answer. This operationalization is however problematic, as it is based on the hypothesis that the subject is capable of critiquing his/her own understanding or has the same definition of understanding as the researcher (he could think he understands what was given to him/her even if the researcher would not agree) and that the subject would not lie (which could be the case, for instance out of embarrassment). To avoid this issue, a survey containing questions on the information could be handed to him/her. However, it is not obvious such a survey would test the relevant information on the experiment. To limit this issue as much as possible, again the questionnaire could be handed to a third-party commission for evaluation, accounting for all the issues stated in the Informed paragraph.

Further issues in decision-capability may concern the actual cognitive abilities of the participants in making decisions. To test this, a psychological screening could be issued, which would give a yes/no answer on the cognitive compatibility of the individual with the experiment. Such an operationalization is based on the reasonable hypothesis a professional would be able to give an accurate, unbiased account.

It is worth noticing that although the operationalizations proposed above are deemed to be sufficiently robust, they are also quite expensive. This arises the question whether some ethical aspects could be sacrificed in favour of a cheaper one. A consequentialist for instance could be in favour of avoiding using third-party commissions if the experimenters are deemed expert enough to handle all the ethical aspects involved in the study, for instance based on long experience and academic records. For a consequentialist, indeed, such a minor change is unlikely to cause consequences in the ethical judgement of the experiment by public opinion.

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